Multi-Modal Sentiment Analysis

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1.- Introduction-

1.1 Decisions to be impacted

This project is designed to support three key decisions in multimodal sentiment analysis. First, it informs model architecture selection, specifically whether sequential models such as LSTMs or more attention-based models are better suited for temporally aligned multimodal sentiment regression. Second, it supports fusion strategy selection, helping determine which fusion mechanism—Early Fusion, Gated Fusion, Cross-modal Attention, or Tensor Fusion—offers the best trade-off between prediction performance, robustness, and model complexity. Third, it informs the decision of whether multimodal modeling is worth the additional cost relative to unimodal baselines, especially in settings where collecting and processing audio and visual signals requires more resources than text alone.

1.2 Business value

On the business side, accurate multimodal sentiment recognition models can significantly enhance the intelligence of human-centered interactive systems. This has key applications in the following areas:

- **Customer Service and Opinion Monitoring**: By jointly analyzing speech, text, and facial expressions, systems can monitor emotional fluctuations in real time and detect potential negative experiences early, thereby improving customer retention and service quality.
- **Intelligent Recommendation and Marketing Optimization**: Emotion-aware models can increase the contextual relevance of advertisements and content recommendations, enabling more precise user profiling and adaptive marketing strategies.
- **Affective Computing and Enhanced Human-Computer Interaction**: In domains such as social robotics, virtual assistants, and educational technology, systems capable of recognizing emotions can interact with users more naturally and empathetically, greatly improving user engagement and satisfaction.

1.3 Why do we care about this project?

We care about this project for both research and practical reasons. From a research perspective, multimodal learning is an important and active area in machine learning, and sentiment intensity prediction is a particularly challenging task because it requires modeling subtle emotional variation rather than only positive-versus-negative polarity. From a practical perspective, real human communication is inherently multimodal. As described in our first report, a speaker may produce neutral words while their tone and facial expression reveal a very different emotional state. This makes sentiment prediction an ideal problem for studying how textual, acoustic, and visual information complement each other. Our project is therefore motivated by a desire to understand not only which model performs best, but also why multimodal fusion helps, when it helps, and how much improvement it can provide over strong unimodal baselines.

1.4 Data assets

The primary data asset used in this project is the CMU-MOSEI dataset, a large-scale multimodal benchmark for sentiment and emotion analysis. CMU-MOSEI contains over 65 hours of annotated video from more than 1,000 speakers and over 23,000 sentence-level segments, with text, audio, and visual modalities temporally aligned at the utterance level. The dataset provides continuous sentiment labels ranging from -3 to +3, which makes it especially suitable for our regression-based formulation of sentiment intensity prediction.

2. -Data Preprocessing-

2.1 Raw Data Structure

The original dataset was provided in a pickle file with three predefined splits: training, validation, and test. Each split contains five fields: text, audio, vision, labels, and id. The three modalities are already aligned to a fixed sequence length of 50 time steps, with shapes $(N, 50, 300)$ for text, $(N, 50, 74)$ for audio, and $(N, 50, 35)$ for vision. The labels have shape $(N, 1, 1)$ and correspond to continuous sentiment scores, while id is used only for sample identification.

Since the modalities were already aligned, no additional feature extraction or synchronization was required. The preprocessing stage therefore focused on cleaning, anomaly analysis, and converting the data into a model-ready format.

2.2 Data Cleaning

Our preprocessing strategy was lightweight and focused on preserving the original temporal structure. We first verified that all three splits contained the required fields and that the numbers of samples were consistent across text, audio, vision, labels, and id. We also confirmed that all modalities shared the same aligned sequence length of 50 time steps.

We then cleaned the feature tensors by replacing any NaN or Inf values with zero, while keeping the labels unchanged. In addition, all modality features were converted from float64 to float32 to reduce memory usage and improve compatibility with deep learning frameworks, and the labels were reshaped from $(N, 1, 1)$ to $(N,)$ for regression training. Finally, because the three modalities have different scales and dimensionalities, we normalized text, audio, and vision separately using the mean and standard deviation computed only from the training split, and applied the same statistics to the validation and test splits to avoid data leakage.

2.3 Outlier Detection

2.3.1 Isolation Forest for Sample-level Outlier Detection

We applied Isolation Forest to detect atypical samples in the multimodal feature space. This method isolates anomalies through recursive random partitioning, where samples with shorter average path lengths are considered more anomalous. Using the concatenated multimodal representation and a contamination setting of 0.05, the model flagged about 5% of the training samples as outliers.

However, this ratio should not be interpreted as the true proportion of corrupted data. Instead, it represents the most atypical subset under the chosen setting.

2.3.2 Temporal Instability and Boundary Effects

We also examined the temporal mean and temporal standard deviation of each modality across the 50 aligned time steps. The results show that all three modalities become less stable in the later part of the sequence, with more noticeable changes after approximately time step 40.

This pattern is likely caused by boundary effects introduced when variable-length original sequences were aligned or truncated to a fixed length of 50. Among the three modalities, audio shows the strongest late-stage fluctuation, suggesting that it is more sensitive to local instability and alignment artifacts.

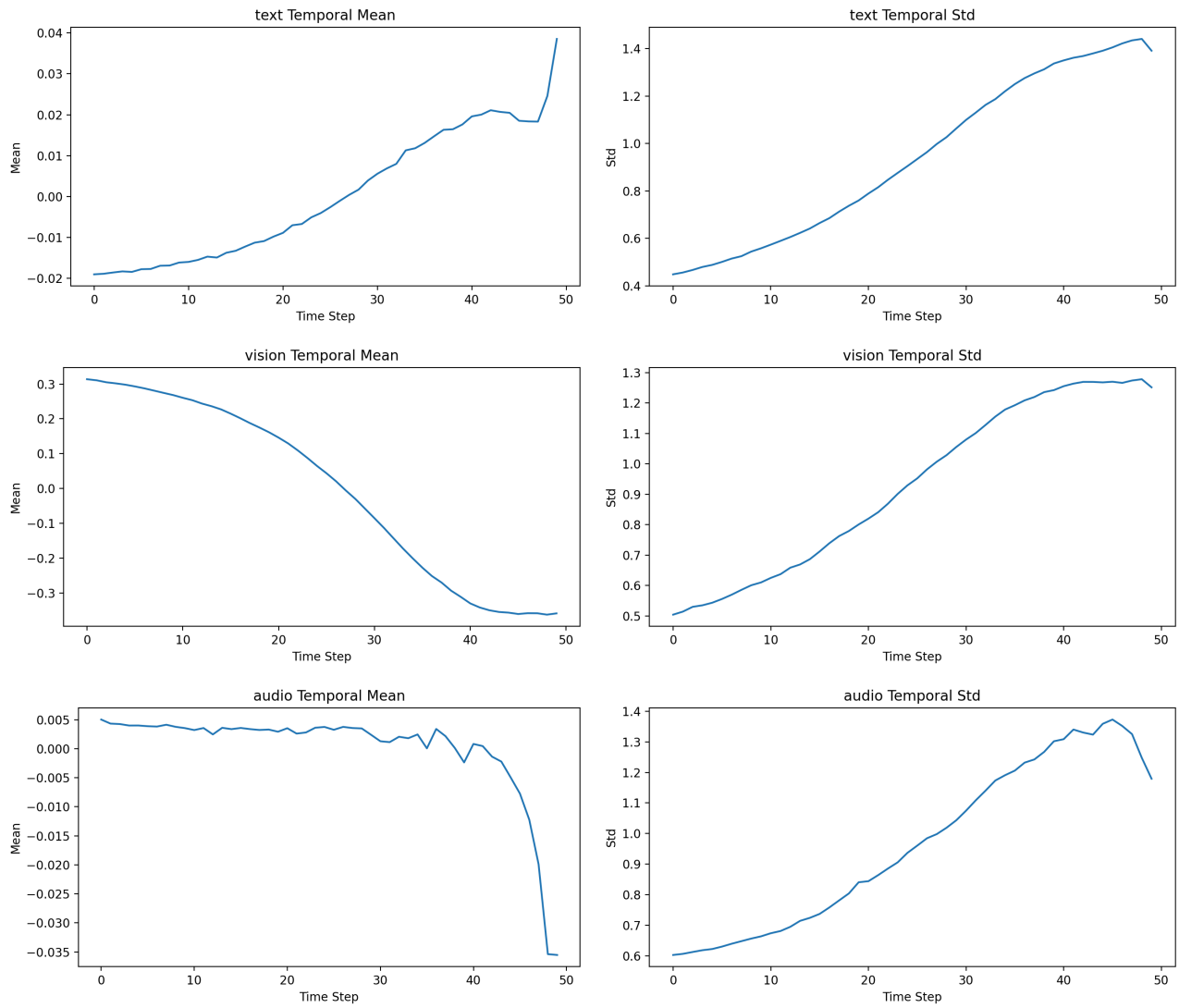

Figure 2.1 Temporal mean and std of text, vision, and audio modalities

2.3.3 Interpretation and Handling Strategy

To better understand the flagged outliers, we analyzed the relationship between Isolation Forest anomaly scores and several dataset characteristics. The correlation between anomaly score and label magnitude was very weak ($r \approx -0.047$), while the correlations with temporal jump intensity were much stronger: about -0.928 for text, -0.854 for audio, and -0.684 for vision.

These results suggest that the detected outliers are mainly associated with temporal instability rather than extreme sentiment labels. Therefore, we did not remove them from the dataset. Instead, we retained all samples and used the analysis to guide downstream model design, especially the use of global sequence aggregation rather than relying only on the final time step.

2.4 Data Encoding for Model Training

After preprocessing, the dataset was organized into a model-ready format:

- text: $(N, 50, 300)$
- audio: $(N, 50, 74)$
- vision: $(N, 50, 35)$
- labels: $(N,)$

This representation preserves the aligned sequence structure and can be directly used for both unimodal and multimodal experiments. No pooling, flattening, or fusion was performed during preprocessing, which keeps the data flexible for later comparisons across LSTM- and Transformer-based models.

2.5 Label Distribution and Task Implication

We also examined the distribution of sentiment labels. Most labels are concentrated around zero, while strongly positive and strongly negative samples are relatively rare. This indicates that the regression target space is imbalanced, with moderate sentiment values appearing much more frequently than extreme cases.

This property suggests that downstream evaluation should include multiple regression metrics, such as MAE, RMSE, and correlation, rather than relying on a single measure.

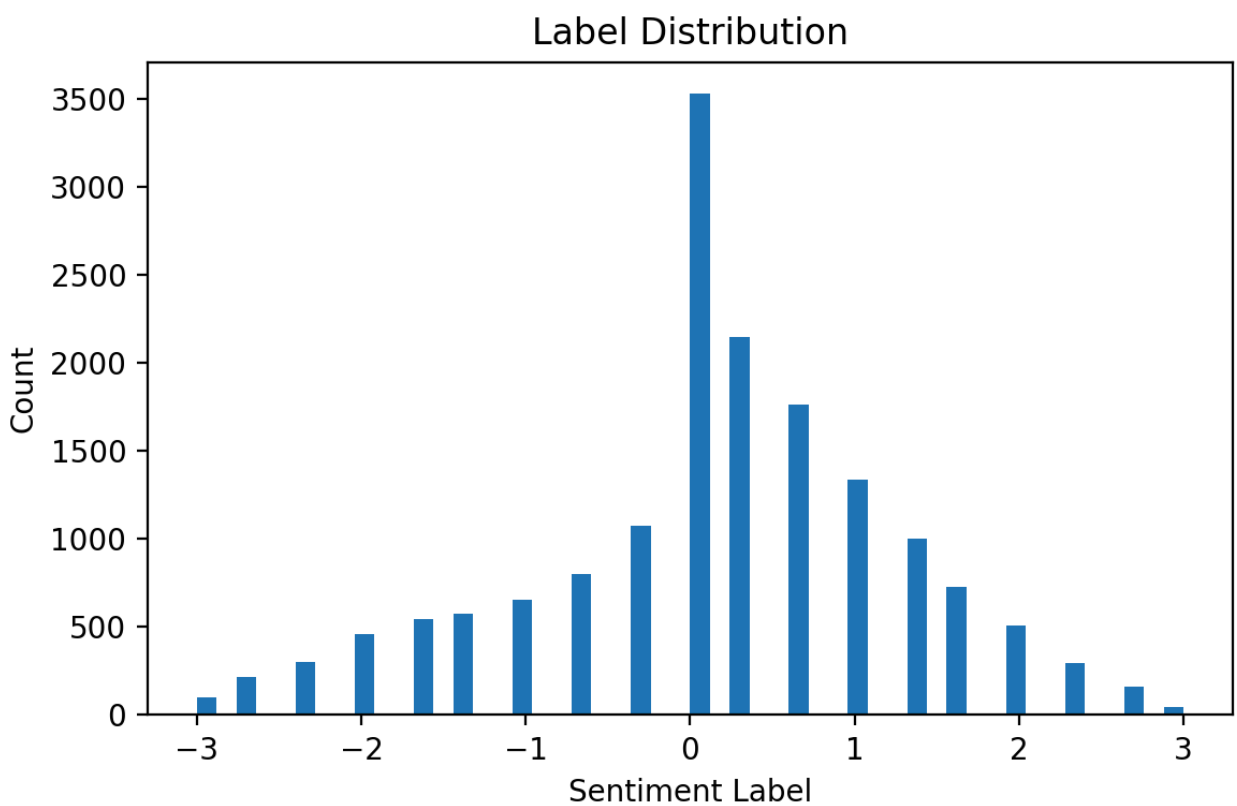

Figure 2.2 Label distribution of sentiment scores

2.6 Summary

In summary, the preprocessing pipeline preserved the original aligned sequence structure while performing essential cleaning, normalization, and anomaly analysis. The results show that the main source of irregularity in the dataset is temporal instability, especially near the later time steps and particularly in the audio modality. After preprocessing, the data are stored in a clean and model-ready sequential format suitable for both LSTM- and Transformer-based experiments.

3.- Model Updates-

3.1 Overall modeling strategy

Our project focuses on multimodal sentiment regression using temporally aligned text, audio, and vision features from the CMU-MOSEI dataset. To support this task, we study two major modeling dimensions simultaneously:

1. Sequence modeling architecture

- LSTM-based models
- Transformer-based models

2. Multimodal fusion strategy

- Early Fusion
- Gated Fusion
- Cross-modal Attention
- Tensor Fusion

Our approach first focuses on identifying the best-performing multimodal architecture, and then uses that model framework to compare multimodal and single-modality inputs in a more controlled way.

3.2 Sequence modeling architectures

To capture temporal dependencies in aligned multimodal sequences, we explored two sequence modeling architectures: LSTM and Transformer.

LSTM

LSTM is well suited for temporally aligned data because it processes sequential information step by step and preserves contextual information over time. In our implementation, the LSTM encoder produces hidden states for all time steps rather than relying only on the final hidden state. We then apply temporal mean pooling to obtain a global sequence representation. This helps preserve sentiment cues distributed across the full utterance.

Transformer

Transformer is also suitable for aligned multimodal sequences because it models temporal dependencies through self-attention, allowing each time step to directly attend to all other positions in the sequence. This is particularly useful when sentiment information is distributed unevenly across an utterance or when important cues appear far apart in time. In our implementation, each modality is first projected into a shared hidden space before being passed through the Transformer encoder. Positional encoding is added to preserve temporal order, since the attention mechanism itself does not encode sequence position.

3.3 Multimodal fusion strategies

We implemented four multimodal fusion strategies and combined them with our sequence modeling architectures.

(a) Early Fusion

In Early Fusion, the text, audio, and vision features are combined at the input level before sequence encoding. This serves as the most direct and simple multimodal baseline. The main advantage of this strategy is simplicity, but it does not explicitly model modality-specific importance or inter-modal relationships.

(b) Gated Fusion

In Gated Fusion, the model learns adaptive weights for the modality representations before combining them. This allows the model to assign different importance to each modality depending on the input sample. The motivation for this strategy is that different utterances may rely on different modalities for sentiment expression.

(c) Cross-modal Attention

In Cross-modal Attention, attention mechanisms are used to explicitly model dependencies across modalities. This allows one modality to focus on the most relevant information from another modality. In sentiment analysis, this is especially useful because text often provides semantic content, while audio and vision provide emotional context.

(d) Tensor Fusion

Tensor Fusion models higher-order interactions among modalities by explicitly constructing cross-modal combinations. This strategy is more expressive than simple concatenation, since it captures unimodal, bimodal, and trimodal relationships. However, it is also computationally heavier and more sensitive to dimensionality and regularization choices.

3.4 Model selection strategy

We compare the multimodal models created by combining our sequence backbones with the four fusion strategies. The goal of this stage is to determine which multimodal architecture provides the strongest sentiment regression performance.

After identifying the best-performing multimodal model, we use that same overall model framework as a reference architecture and train it on single modalities separately. In other words, we apply the strongest model design to:

- text-only input
- audio-only input
- vision-only input

and then compare those results against the full multimodal version.

3.5 Modality projection

One challenge in this project is that the three modalities have very different raw feature dimensions:

- Text: 300
- Audio: 74
- Vision: 35

This dimensional imbalance may cause the text modality to dominate the shared representation space during multimodal learning. To address this, we introduced modality-specific projection layers that map each modality into a shared latent dimension before fusion and sequence encoding.

3.6 Performance metrics

Because this is a regression task, we evaluate all models using:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- Pearson Correlation

MAE and RMSE measure prediction error, where lower values indicate better performance. Pearson correlation measures the agreement between predicted sentiment scores and ground-truth labels, where higher values indicate better alignment.

3.7 Machine Learning Morphism (MLM) workflow

Our current machine learning workflow can be summarized as:

Aligned multimodal input features (text, audio, vision)

1. Data Cleaning and Preprocessing
2. Modality Projection and Normalization
3. Sequence Encoding (LSTM or Transformer)
4. Multimodal Fusion (Early Fusion, Gated Fusion, Cross-modal Attention, or Tensor Fusion)
5. Temporal Aggregation and Task-Specific Representation
6. Regression Head
7. Continuous Sentiment Prediction
8. Evaluation Using MAE, RMSE, and Pearson Correlation

This workflow provides a unified framework for comparing multiple architectures and fusion strategies under the same multimodal sentiment regression task.

4.- Next Steps-

Week 1-2. Hyperparameter Tuning and Model Refinement

We will further tune the main hyperparameters of our models, including hidden size, learning rate, dropout, number of layers, and projection dimension. The goal is to determine whether additional performance gains can be achieved through more systematic optimization of the current architectures and to identify the strongest model configuration.

Week 3. Model Comparison

During this stage, we will also further compare LSTM-based and Transformer-based models under matched fusion settings to better understand whether improvements come mainly from the sequence encoder, the fusion strategy, or their interaction.

Week 4. Finalize Visualizations, Results Interpretation, and Report Preparation

We will refine the clarity and presentation quality of our figures, result tables, and workflow diagrams so that the final report communicates the methodology and findings more effectively. In addition, we plan to conduct error analysis on representative cases to better understand where multimodal fusion helps most and where the models still struggle.

Github: https://github.com/Wakamo-0716/ESE-527-Project.git